Ali Keramati

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Education

University of California Irvine **Sep. 2024 – Jun. 2029**

Ph.D in **Education Data Science** (GPA: 4/4) **Expected Graduation Date: June 2029** *Irvine, CA, USA*

- **Relevant Courses:** *Neural Networks and Deep Learning, Deep Generative Models, Neural Architecture of Language, Language Models in Cognitive Science, Learning Analytics Practicum, Qualitative Research Methods, Research Epistemologies and Methods, Applied Regression, Brain Inspired Learning Algorithms*

University of Tehran **Sep. 2019 – Jul. 2024**

B.Sc. in **Computer Engineering** (GPA: 18.51/20) + Minor in **Psychology and Education** (GPA: 19.21/20) *Tehran, Iran*

- **Major Degree Courses:** *Discrete Mathematics, Data Structures and Algorithms, Artificial Intelligence, Machines and Language Theory, Engineering Probability and Statistics, Internet Engineering, Foundations of IT, Large Language Models*
- **Minor Degree Courses:** *Educational Psychology, Instructional Design, Curriculum Development, Intro to Psychological Tests, Teaching Methods, Qualitative and Quantitative Research Methods, Philosophical Schools and Educational Theories*

Research Interests

- **AI Harness**: Tool-use/MCP/RAG/Memory/Reasoning
- **HCI**: Human-Centered AI/Human-in-the-Loop AI
- **Online/Offline Evaluation**: LLM-as-Judge, Reliability
- **Conversational AI & TOD Systems**: SAT/DSAT
- **Fine-Tuning** & Alignment: LoRA/Adapters/RLxF
- **RL**: MDPs/Reward Modeling/Constrained/Actor–Critic
- Explainable, Trustworthy, and Responsible AI
- **Multi-Agent Systems**: Debate/Collaborate
- **User Simulation**: Personas, Behavioral Modeling
- Synthetic Data Generation/Quality Assurance/validation

Work Experience

Visiting Researcher at Harvard University **Jun. 2025 - Sep. 2025**

Child-Centered AI Lab; PI: Prof. Y. Xu *Cambridge, MA*

- Developed a LangGraph-based science chatbot using graph-based dialogue and state management for transparency and explainability **(XAI)**, with child-safety safeguards supporting **Trustworthy AI**. Aligned the system through a **Constitutional AI**-grounded multi-agent self-correction pipeline and validated it against K–12 standards using multiple LLM judges.
- Built a hybrid **child-speech ASR** pipeline (ages 3–5) that fuses **GPT-Transcribe**, a child-speech Qwen3-ASR model, a WavLM-CTC phoneme (IPA) recognizer, and dialogue context via LLM reconstruction, preserving child grammar and flagging uncertain words; designed a leave-one-child-out WER ablation.

Applied Scientist Intern @ Microsoft **Jun. 2026 – Sep. 2026**

Copilot Studio, Evals Team; Manager: Tommy Guy *Redmond, WA*

- Developed evaluation, user-simulation, and RL pipelines for multi-turn agents in Copilot Studio, combining LLM-as-judge evaluators, persona-driven synthetic users, and sandbox-derived rewards to evaluate agent reliability and support PPO-based post-training across stateful task-oriented workflows.
- Co-developed and open-sourced **ThinkingBox** and **ThinkingBox-Bench**, an MCP-compatible agent sandbox and 507-task benchmark for evaluating policy adherence, tool use correctness, and task-completion reliability in stateful business workflows.

Selected Publications

[1]Li Z.†, Ko Y.†, **Keramati A.**†, et al. **"One Success Isn't Reliability: Thinkingbox, a Sandbox and Benchmark for Agents in Stateful Business Workflows."** *Work completed during a Microsoft internship.*

[2]**Keramati A.**, Zhou S., Mehrotra S., and Warschauer M. **"MADRAG: Multi-Agent Debate with Retrieval-Augmented Generation for Training-Free Analytic Essay Scoring."** ACL 2026.

[3]**Keramati A.**, Cheok J., Horne J., and Warschauer M. **"The Confident Liar: Diagnosing Multi-Agent Debate with Log-Probabilities and LLM-as-Judge."** ACL 2026.

[4]**Keramati A.**†, Cao J.†, Mohammadi I., Warschauer M., and Shi Y. **"Simulating Students' Java Programming Errors with Large Language Models."** EDM 2026. Best Short Paper Award.

[5]**Keramati A.**, Cheok J., Horne J., and Warschauer M. **"The First Tokens Matter: Early Confidence Signals for Evaluating LLM Reasoning."** ICLR 2026.

[6]**Keramati A.** and Warschauer M. **"Application of Multi-Agent Systems for Essay Scoring."** NeurIPS 2025.

[7]Tajik E., Borchers C., Shahrokhian B., Simon S., **Keramati A.**, Pal S., and Sankaranarayanan S. **"Disagreement as Data: Reasoning Trace Analytics in Multi-Agent Systems."** LAK 2026.

[8]Mohammadi I., **Keramati A.**, and Rismanchian S. **"Mapping Instructors' Attitudes Towards Cheating with AI: An Exploratory Analysis Using Forum Analysis."** ICLS 2026.

[9]Fallah A., **Keramati A.**, Nazari M. A., and Mirfazeli F. S. **"Automating Theory of Mind Assessment with a LLaMA-3-Powered Chatbot: Enhancing Faux Pas Detection in Autism."** ICCKE 2024.

[10]**Keramati A.** and Nooralahzadeh F. **"Reading Is Not Steering: Auditing Cross-Lingual Chain-of-Thought Probes with Leakage Controls"** *Under Review: ICLR 2027*

Honors & Awards

ACL 2026 — Student Volunteer & Conference Award Recipient **Jul. 2026**
Selected as a Student Volunteer and awarded a competitive conference funding for participation *San Diego, CA*

NeurIPS 2025 — AI Education Workshop Program **Dec. 2025**
Accepted Oral Presentation *San Diego, CA*

Scholarship Recipient, LearnLab Summer School at Carnegie Mellon University **Jul. 2025**
Awarded funding to attend the program on Computer Science Education Research track at CMU *Pittsburgh, PA*

Graduate Ambassador Candidate – TPC Mentor Fellowship at *UC Irvine* **Jun. 2025**
Chosen to oversee mentor–mentee coordination and selection, enhancing communication and program engagement *Irvine, CA*

Top 10 — B.Sc. Computer Engineering (2019 cohort) at *University of Tehran* **Jul. 2024**
Ranked within the top 10 of the graduating cohort based on cumulative GPA *Tehran, Iran*

Technical Skills

Programming Language: Python, C/C++/C#, Java, Golang, JavaScript (Node.js, React), SQL, HTML/CSS, Verilog
Agentic Frameworks: LangChain, LangGraph, AutoGen, CrewAI, Swarm (OpenAI), Mem0AI, ReAct, Haystack
DL Fine-Tuning & Alignment: PyTorch, Hugging Face Transformers, PEFT, TRL, OpenDelta, Diffusers, vLLM
Retrieval & Memory: FAISS, ChromaDB, Neo4j, Weaviate, Elasticsearch, Redis, PostgreSQL, MongoDB
Data Analytics: Pandas, NumPy, Scikit-learn, Spark, R, Power BI, LearnSphere, OpenLA, PyBKT, ATLAS.ti, Dedoose
Human-Centered & Cognitive Tools: Unity ML-Agents, PsychoPy, PsyToolkit, OpenAI Agents SDK

Other Research Projects

University of California, Irvine

* *PapyrusAI:* Designed a personalized generative writing coach leveraging an **agentic RL** pipeline capable of autonomous decision-making for tool invocation, web search, and dynamic **RAG** utilization. Integrated fine-tuned **BERT** for labeling argumentative essays and LLaMA-3.1-8B with **LoRA** + 4-bit **quantization** for essay evaluation.
* Investigating uncertainty localization in agentic AI systems, enabling agents to detect errors in SQL/Python outputs, optimize computational and human-interaction costs, and decide when to halt, resample, or request clarification.
* Developed an adaptive **few-shot RAG** system with continual memory curation: admission control, candidate→gold promotion, and feedback-driven ranking to keep high-quality exemplars highly retrievable as the database grows.
* As **TA**, developed an **automated grading** and **feedback** system for weekly assignments and co-built an **AI detection RL system** using LLMs to identify AI-generated submissions based on writing patterns, timing, and key concepts.
* Extending VAEs for Collaborative Filtering with modern training strategies on large implicit-feedback datasets, and benchmarking these VAE-based recommenders against language-model–based recommenders trained on the same data.

LearnLab Summer School (2025) at Carnegie Mellon University

* Developed a multi-agent **student simulator + verifier** framework for generating and verifying Java programs with realistic logical errors. Designed a student simulator using *Self-Refine* prompting, and a verifier agent implementing an **RLAIF** loop for structured error assessment and refinement. Evaluated performance via AST- and ZSS-based metrics.
* Analyzing Reddit posts from *r/Teachers* using **LDA** topic modeling, NLP-based classification, and qualitative coding to examine teachers' use of AI tools; training an LLM-based **classifier** to generalize coded themes across large datasets.

University of Tehran

* Developed a multi-agentic chatbot for open learner modeling, using a multi-stage conversational flow to dynamically **profile** and update learner characteristics based on their in-chat interactions. Integrated a **RAG** system that uses a learner's profile as a query to retrieve the top 3 relevant profiles, aligning personalized activity recommendations.
* Automated generation of interactive learning modules from **IEEE LOM metadata** and integrated Avida-ED simulations; applied genetic algorithms to personalize activity parameters and difficulty levels.
* Implemented Theory of Mind assessment for individuals with Autism by developing a **LLaMA-3 chatbot** to administer the Faux Pas Recognition Test. The system uses a multi-phase pipeline that features interactive rehabilitation dialogues with adaptive hints and an automated scoring phase using an **LLM-as-judge** to generate clinician-ready reports.

Certifications

Generative AI with Large Language Models **Jul. 2022**
DeepLearning.AI — Coursera *Online*

Fundamentals of Reinforcement Learning **Aug. 2023**
University of Alberta — Coursera *Online*

LangChain: Chat with Your Data **Jul. 2024**
DeepLearning.AI *Online*

Microsoft Certified: Azure AI Fundamentals **Oct. 2025**
Microsoft Learn *Online*